Answer the questions in the spaces provided on the question sheets. If you run out of room for an answer, continue on the back of the page.

Name: _____

Student ID: _____

Quiz 4 - 09/17 Content

MATH 2263

09/22/2026

This quiz is worth 5 points and consists of 1 questions. You have 5 minutes to complete the quiz.

Question	Points
1	5
Total:	5

1. (5 points) Use the Squeeze Theorem to determine the value of

$$\lim_{x\to 0} x\cos\left(\frac{1}{x}\right).$$

Solution: We want to apply the squeeze theorem, which says that for function $f(x), g(x), h(x)$ so that:

1. $g(x) \leq f(x) \leq h(x)$ around $x = a$
2. $\lim_{x\to a} g(x) = \lim_{x\to a} h(x)$

we have $\lim_{x\to a} f(x) = \lim_{x\to a} g(x) = \lim_{x\to a} h(x)$.

We then want to bound $f(x) = x\cos\left(\frac{1}{x}\right)$ by functions whose limits at $x = 0$ are equal. Since the range of cos is $[-1, 1]$, we know that

$$x \cdot (-1) \leq x \cdot \cos\left(\frac{1}{x}\right) \leq x \cdot 1 \tag{1}$$

and since $\lim_{x\to 0} x = \lim_{x\to 0} -x = 0$ we get that

$$\lim_{x\to 0} x\cos\left(\frac{1}{x}\right) = 0 \tag{2}$$